

HOWTO Load a Dataset into a Data Frame in a Remote Server

1. Overview

The following line of Python loads a data set, say “Iris.csv” into a Pandas data frame df.

```
df = 'DATA/Iris.csv'
```

- It is assumed that this line of code is executed in a remote server in the cloud.
- This statement loads the data set saved in the data file Iris.csv into the Pandas data frame df.
- This data file is saved under a sub-folder named “DATA” that is located in the same folder where the Jupyter Notebook document *.ipynb is found.

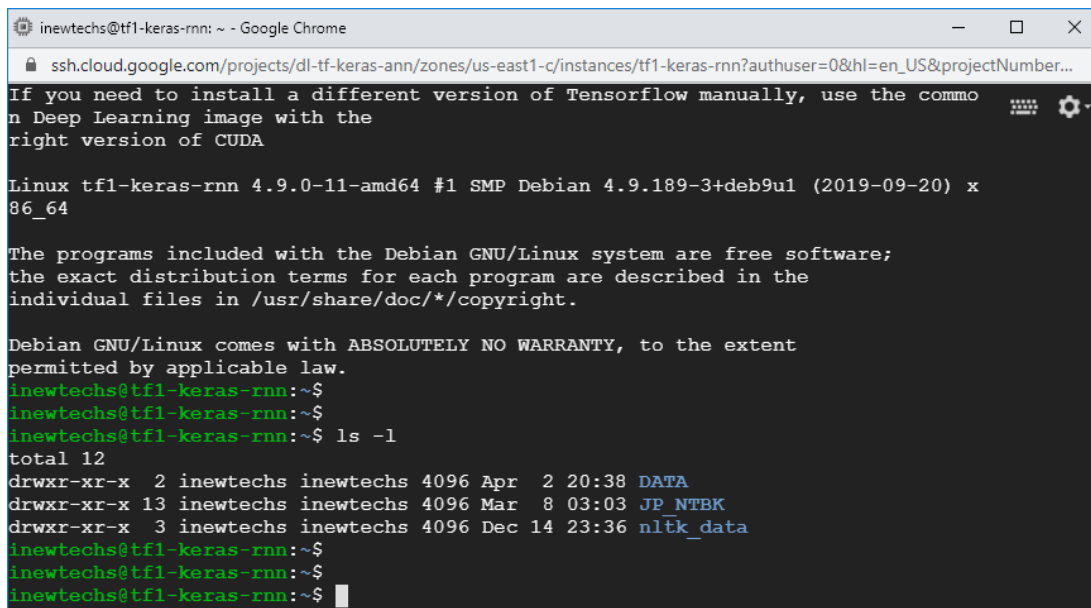
2. HOWTO Load a Dataset into a Dataframe in a Remote Server

IMPOTANT NOTES:

--> It is assumed that Jupyter Notebook server has started in the folder ~/JP_NTBK that is found under the home directory of the user.

--> It means that all the Jupyter Notebook documents can be found in this folder

2.1 Open an SSH to the Remote Server



```
inewtechs@tf1-keras-rnn: ~ - Google Chrome
ssh.cloud.google.com/projects/dl-tf-keras-ann/zones/us-east1-c/instances/tf1-keras-rnn?authuser=0&hl=en_US&projectNumber...
If you need to install a different version of Tensorflow manually, use the commo
n Deep Learning image with the
right version of CUDA

Linux tf1-keras-rnn 4.9.0-11-amd64 #1 SMP Debian 4.9.189-3+deb9u1 (2019-09-20) x
86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$ ls -l
total 12
drwxr-xr-x  2 inewtechs inewtechs 4096 Apr  2 20:38 DATA
drwxr-xr-x 13 inewtechs inewtechs 4096 Mar  8 03:03 JP_NTBK
drwxr-xr-x  3 inewtechs inewtechs 4096 Dec 14 23:36 nltk_data
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$
```

The terminal shows the user’s home directory in the remote server

2.2 Create a sub-folder named DATA under the folder JP_NTBK

- Access the folder **JP_NTBK**
 - Run the command line: **cd JP_NTBK**

```
inewtechs@tf1-keras-rnn: ~/JP_NTBK - Google Chrome
ssh.cloud.google.com/projects/dl-tf-keras-ann/zones/us-east1-c/instances/tf1-keras-rnn?authuser=0&hl=en_US&projectNumber=232743561122
If you need to install a different version of Tensorflow manually, use the commo
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right version of CUDA

Linux tf1-keras-rnn 4.9.0-11-amd64 #1 SMP Debian 4.9.189-3+deb9u1 (2019-09-20) x
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inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$ ls -l
total 12
drwxr-xr-x  2 inewtechs inewtechs 4096 Apr  2 20:38 DATA
drwxr-xr-x 14 inewtechs inewtechs 4096 Apr 11 03:38 JP_NTBK
drwxr-xr-x  3 inewtechs inewtechs 4096 Dec 14 23:36 nltk_data
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$ cd JP_NTBK
inewtechs@tf1-keras-rnn:~/JP_NTBK$
inewtechs@tf1-keras-rnn:~/JP_NTBK$
inewtechs@tf1-keras-rnn:~/JP_NTBK$
```

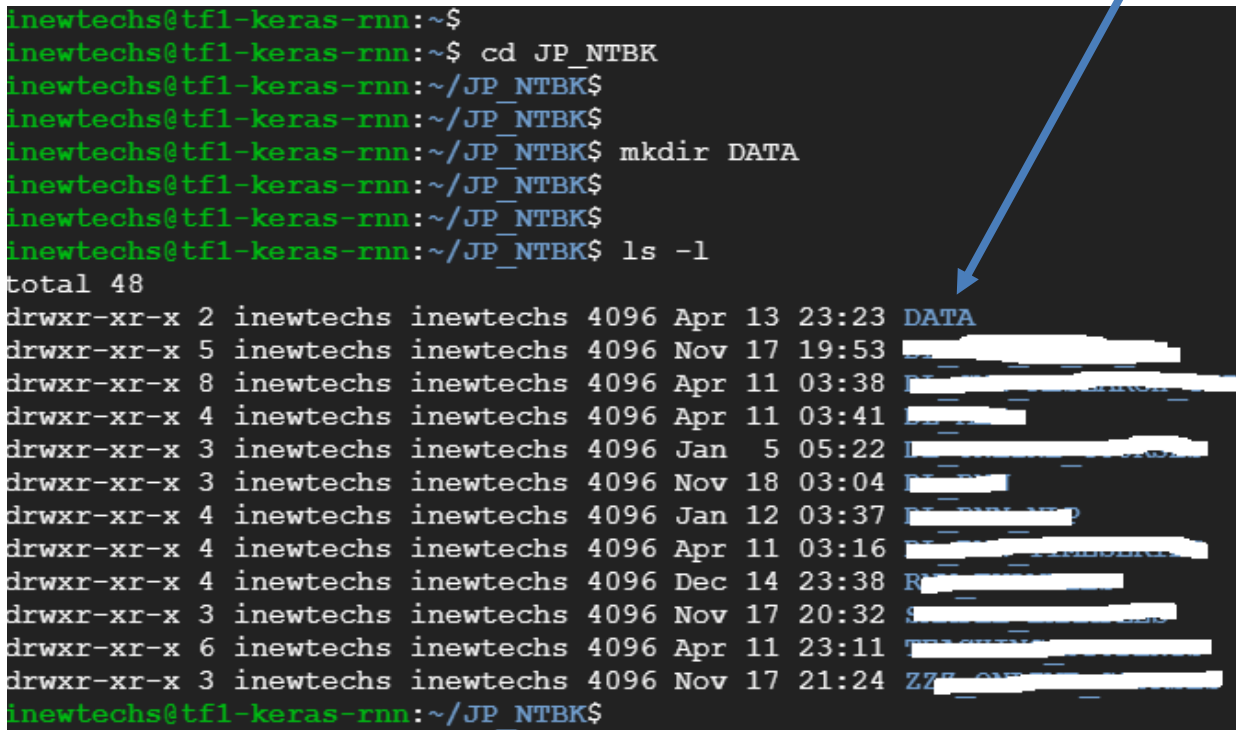
- Create a **new sub-folder** named “**DATA**” under the folder JP_NTBK
 - Run the command line: **mkdir DATA**

```
inewtechs@tf1-keras-rnn: ~/JP_NTBK - Google Chrome
ssh.cloud.google.com/projects/dl-tf-keras-ann/zones/us-east1-c/instances/tf1-keras-rnn?authuser=0&hl=en_US&projectNumber=232743561122
Linux tf1-keras-rnn 4.9.0-11-amd64 #1 SMP Debian 4.9.189-3+deb9u1 (2019-09-20) x
86_64

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inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$ ls -l
total 12
drwxr-xr-x  2 inewtechs inewtechs 4096 Apr  2 20:38 DATA
drwxr-xr-x 14 inewtechs inewtechs 4096 Apr 11 03:38 JP_NTBK
drwxr-xr-x  3 inewtechs inewtechs 4096 Dec 14 23:36 nltk_data
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$ cd JP_NTBK
inewtechs@tf1-keras-rnn:~/JP_NTBK$
inewtechs@tf1-keras-rnn:~/JP_NTBK$
inewtechs@tf1-keras-rnn:~/JP_NTBK$ mkdir DATA
inewtechs@tf1-keras-rnn:~/JP_NTBK$
inewtechs@tf1-keras-rnn:~/JP_NTBK$
inewtechs@tf1-keras-rnn:~/JP_NTBK$
```

- Verify that the new sub-folder has been created
 - Run the command line: **ls -l**



```

inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$ cd JP_NTBK
inewtechs@tf1-keras-rnn:~/JP_NTBK$
inewtechs@tf1-keras-rnn:~/JP_NTBK$
inewtechs@tf1-keras-rnn:~/JP_NTBK$ mkdir DATA
inewtechs@tf1-keras-rnn:~/JP_NTBK$
inewtechs@tf1-keras-rnn:~/JP_NTBK$
inewtechs@tf1-keras-rnn:~/JP_NTBK$ ls -l
total 48
drwxr-xr-x 2 inewtechs inewtechs 4096 Apr 13 23:23 DATA
drwxr-xr-x 5 inewtechs inewtechs 4096 Nov 17 19:53
drwxr-xr-x 8 inewtechs inewtechs 4096 Apr 11 03:38
drwxr-xr-x 4 inewtechs inewtechs 4096 Apr 11 03:41
drwxr-xr-x 3 inewtechs inewtechs 4096 Jan 5 05:22
drwxr-xr-x 3 inewtechs inewtechs 4096 Nov 18 03:04
drwxr-xr-x 4 inewtechs inewtechs 4096 Jan 12 03:37
drwxr-xr-x 4 inewtechs inewtechs 4096 Apr 11 03:16
drwxr-xr-x 4 inewtechs inewtechs 4096 Dec 14 23:38
drwxr-xr-x 3 inewtechs inewtechs 4096 Nov 17 20:32
drwxr-xr-x 6 inewtechs inewtechs 4096 Apr 11 23:11
drwxr-xr-x 3 inewtechs inewtechs 4096 Nov 17 21:24
inewtechs@tf1-keras-rnn:~/JP_NTBK$

```

IMPOTANT NOTES:

--> It is expected that the new sub-folder DATA should be found inside the folder JP_NTBK

2.3 Change directory back to the home directory

- Change directory back to the home directory
 - Run the command line: **cd**

```

inewtechs@tf1-keras-rnn:~/JP_NTBK$
inewtechs@tf1-keras-rnn:~/JP_NTBK$
inewtechs@tf1-keras-rnn:~/JP_NTBK$ cd
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$

```

2.4 Upload the data file to the remote server

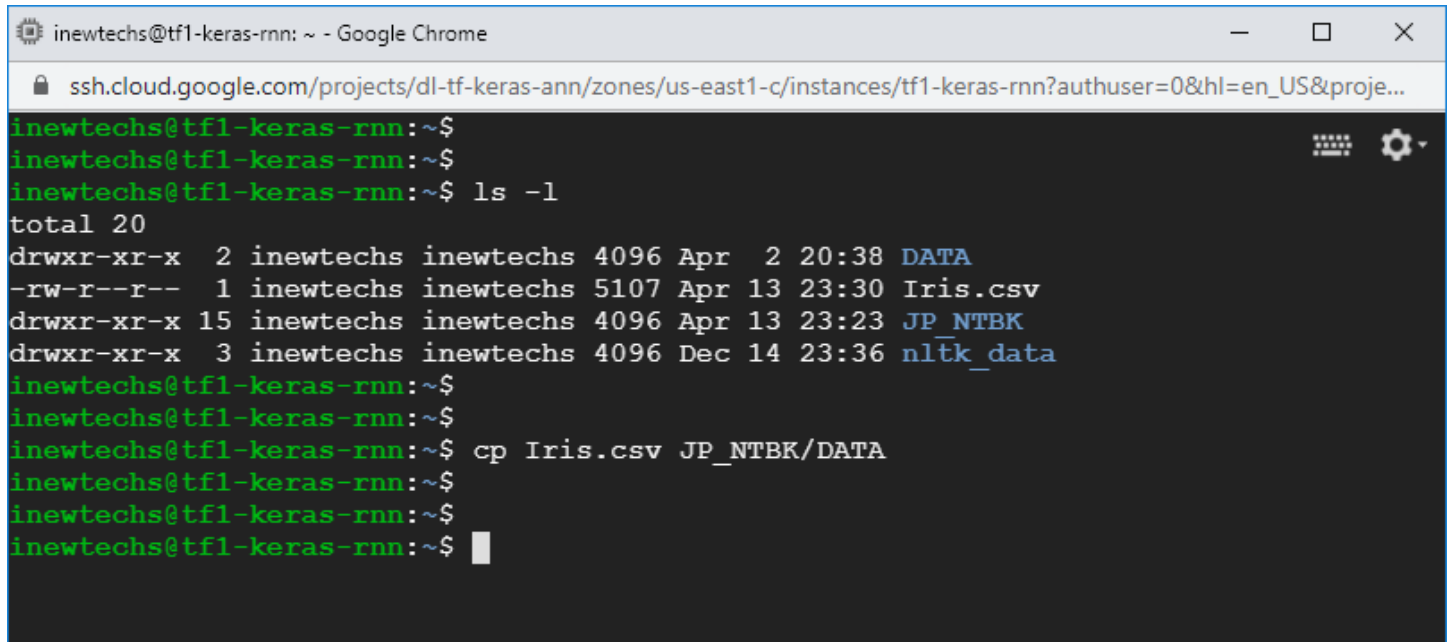
Follow the step discussed in the document:

[HOWTO_upload_files_to_remote_server_using_GCP_SSH.docx](#)

```
inewtechs@tf1-keras-rnn:~$  
inewtechs@tf1-keras-rnn:~$  
inewtechs@tf1-keras-rnn:~$ ls -l  
total 20  
drwxr-xr-x  2 inewtechs inewtechs 4096 Apr  2 20:38 DATA  
-rw-r--r--  1 inewtechs inewtechs 5107 Apr 13 23:30 Iris.csv  
drwxr-xr-x 15 inewtechs inewtechs 4096 Apr 13 23:23 JP_NTBK  
drwxr-xr-x  3 inewtechs inewtechs 4096 Dec 14 23:36 nltk_data  
inewtechs@tf1-keras-rnn:~$
```

2.5 Copy the data file into the sub-folder ~/JP_NTBK/DATA

- Copy the data file into the sub-folder:
 - Run the command line: **cp Iris.csv JP_NTBK/DATA**



```
inewtechs@tf1-keras-rnn: ~ - Google Chrome  
ssh.cloud.google.com/projects/dl-tf-keras-ann/zones/us-east1-c/instances/tf1-keras-rnn?authuser=0&hl=en_US&proje...  
inewtechs@tf1-keras-rnn:~$  
inewtechs@tf1-keras-rnn:~$  
inewtechs@tf1-keras-rnn:~$ ls -l  
total 20  
drwxr-xr-x  2 inewtechs inewtechs 4096 Apr  2 20:38 DATA  
-rw-r--r--  1 inewtechs inewtechs 5107 Apr 13 23:30 Iris.csv  
drwxr-xr-x 15 inewtechs inewtechs 4096 Apr 13 23:23 JP_NTBK  
drwxr-xr-x  3 inewtechs inewtechs 4096 Dec 14 23:36 nltk_data  
inewtechs@tf1-keras-rnn:~$  
inewtechs@tf1-keras-rnn:~$  
inewtechs@tf1-keras-rnn:~$ cp Iris.csv JP_NTBK/DATA  
inewtechs@tf1-keras-rnn:~$  
inewtechs@tf1-keras-rnn:~$  
inewtechs@tf1-keras-rnn:~$
```

2.6 Verify that the data file has been copied into the sub-folder

- Change directory to the sub-folder
 - Run the command line: **cd JP_NTBK/DATA**

```
inewtechs@tf1-keras-rnn: ~/JP_NTBK/DATA - Google Chrome
ssh.cloud.google.com/projects/dl-tf-keras-ann/zones/us-east1-c/instances/tf1-keras-rnn?authuser=0&hl=en_US&proje...
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$ ls -l
total 20
drwxr-xr-x  2 inewtechs inewtechs 4096 Apr  2 20:38 DATA
-rw-r--r--  1 inewtechs inewtechs 5107 Apr 13 23:30 Iris.csv
drwxr-xr-x 15 inewtechs inewtechs 4096 Apr 13 23:23 JP_NTBK
drwxr-xr-x  3 inewtechs inewtechs 4096 Dec 14 23:36 nltk_data
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$ cp Iris.csv JP_NTBK/DATA
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$ cd JP_NTBK/DATA
inewtechs@tf1-keras-rnn:~/JP_NTBK/DATA$
inewtechs@tf1-keras-rnn:~/JP_NTBK/DATA$
inewtechs@tf1-keras-rnn:~/JP_NTBK/DATA$
```

- List the contents of the sub-folder
 - Run the command line: **ls -l**

```
inewtechs@tf1-keras-rnn: ~/JP_NTBK/DATA - Google Chrome
ssh.cloud.google.com/projects/dl-tf-keras-ann/zones/us-east1-c/instances/tf1-keras-rnn?authuser=0&hl=en_US&proje...
drwxr-xr-x  3 inewtechs inewtechs 4096 Dec 14 23:36 nltk_data
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$ cp Iris.csv JP_NTBK/DATA
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$
inewtechs@tf1-keras-rnn:~$ cd JP_NTBK/DATA
inewtechs@tf1-keras-rnn:~/JP_NTBK/DATA$
inewtechs@tf1-keras-rnn:~/JP_NTBK/DATA$
inewtechs@tf1-keras-rnn:~/JP_NTBK/DATA$ ls -l
total 8
-rw-r--r--  1 inewtechs inewtechs 5107 Apr 13 23:40 Iris.csv
inewtechs@tf1-keras-rnn:~/JP_NTBK/DATA$
inewtechs@tf1-keras-rnn:~/JP_NTBK/DATA$
inewtechs@tf1-keras-rnn:~/JP_NTBK/DATA$
inewtechs@tf1-keras-rnn:~/JP_NTBK/DATA$
```